Task 6: Implement Basic Jenkins CI/CD Pipeline

Objective

To configure Jenkins to retrieve source code from GitHub and automatically execute the CI/CD pipeline when changes are detected.

Jenkins Job Configuration

A Jenkins pipeline job named: `devops-ci-cd-pipeline` was created.

The pipeline was configured using:

```
Definition: Pipeline script from SCM
SCM: Git
Branch: */main
Script Path: Jenkinsfile
```

The GitHub repository was provided as the SCM repository.

Pipeline Execution

Jenkins retrieves the Jenkinsfile and source code from GitHub.

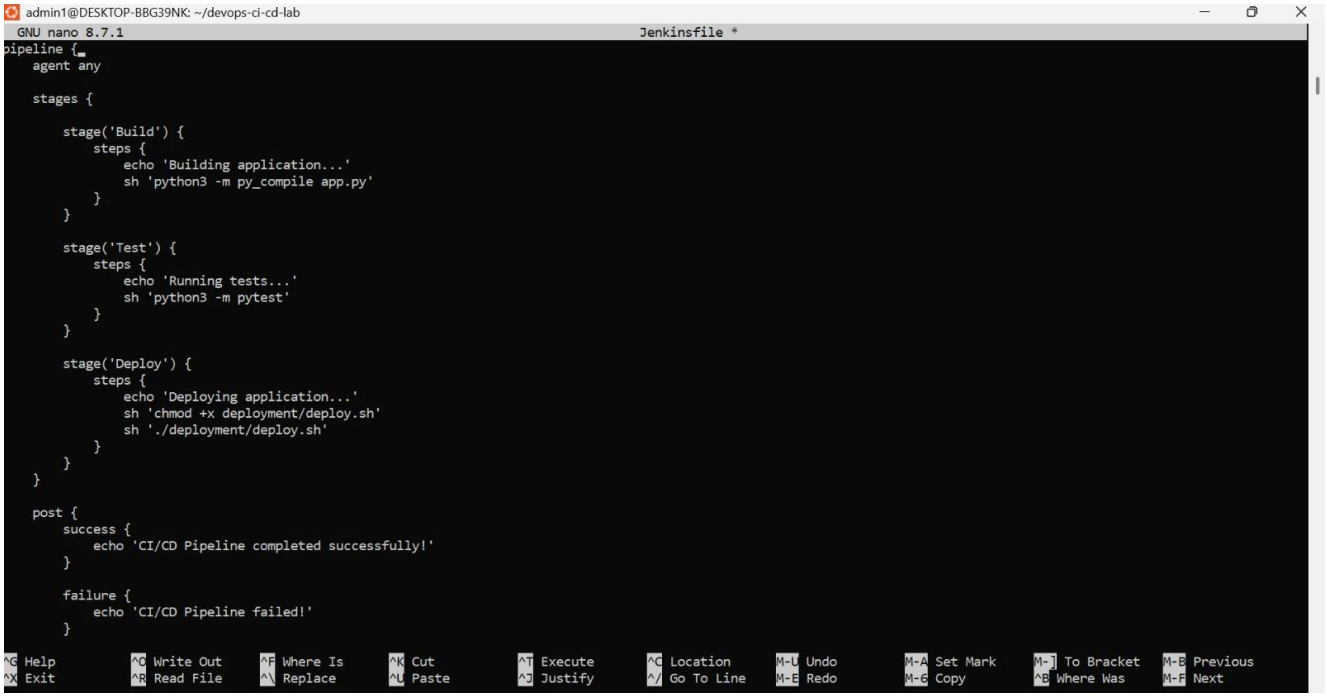
The pipeline then executes:

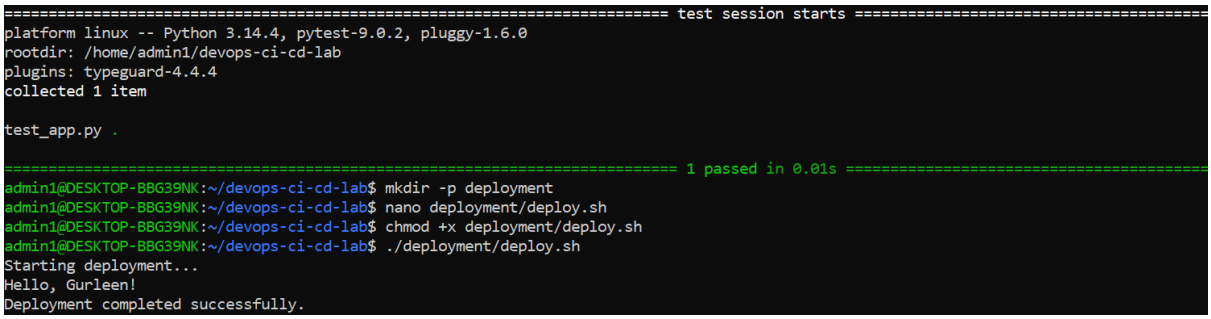
CHECKOUT
↓
BUILD
↓
TEST
↓
DEPLOY

The pipeline successfully completed all stages.

The console output showed:

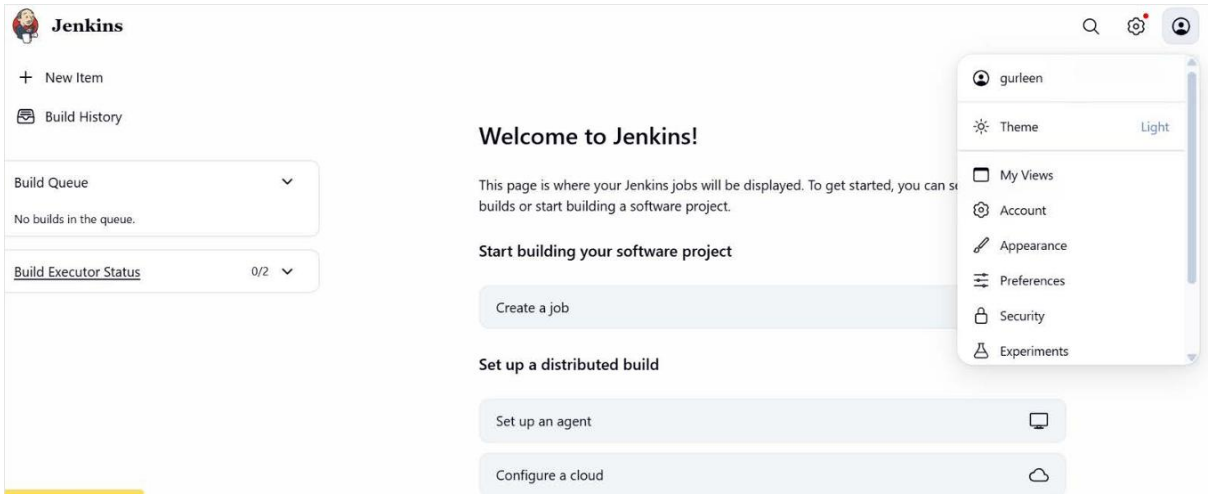
```
1 passed
Deployment completed successfully.
CI/CD Pipeline completed successfully!
Finished: SUCCESS
```

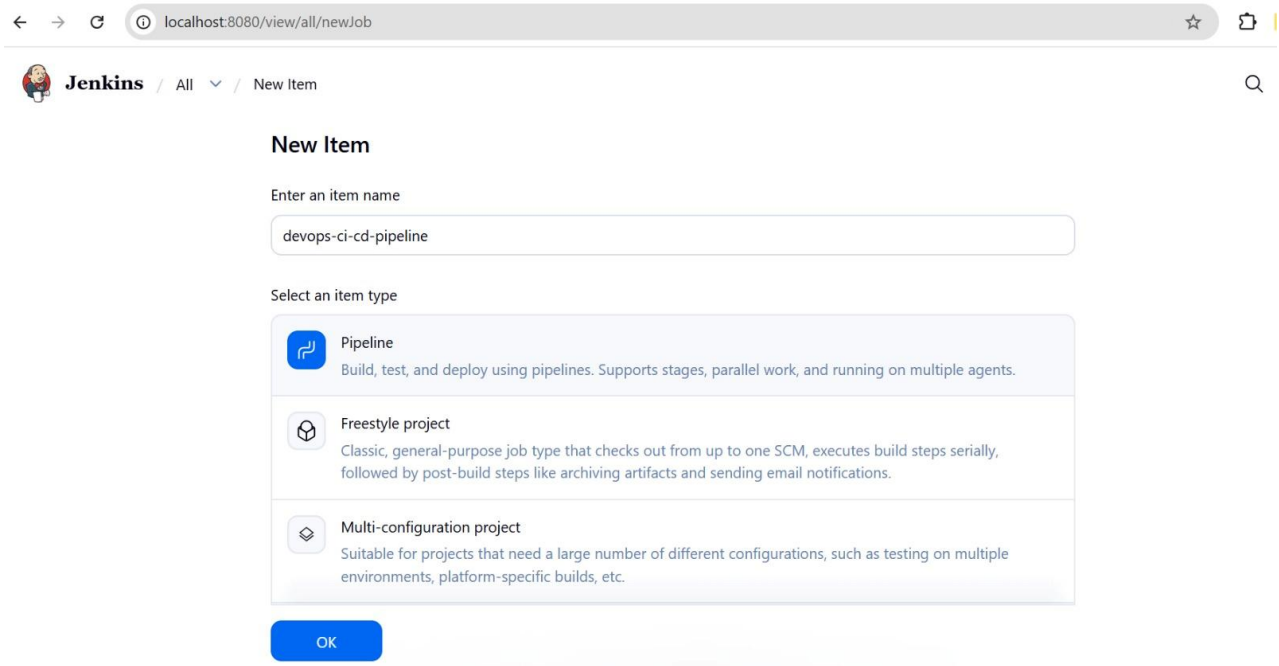


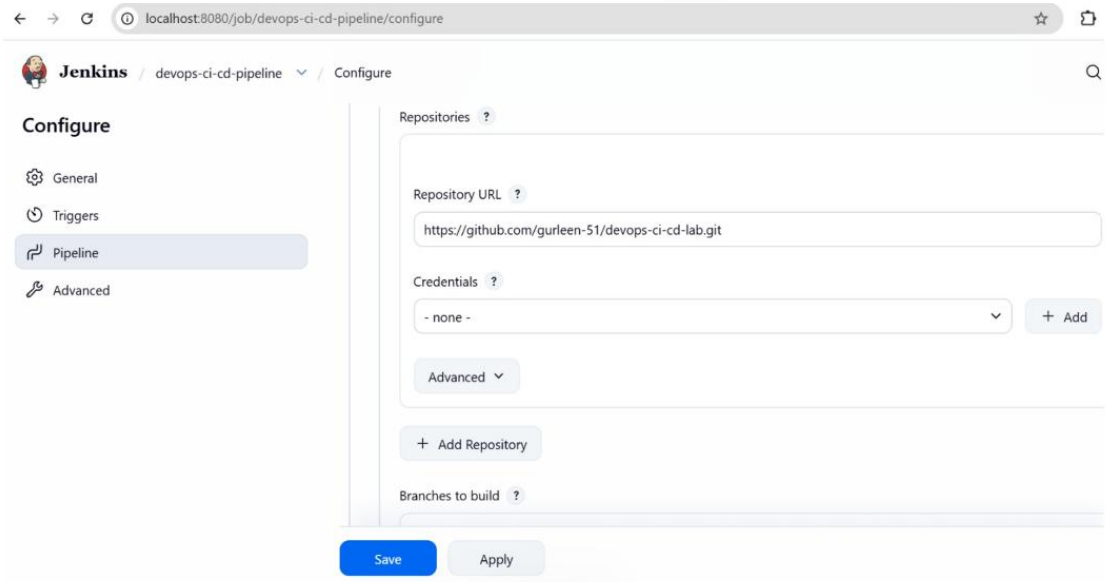


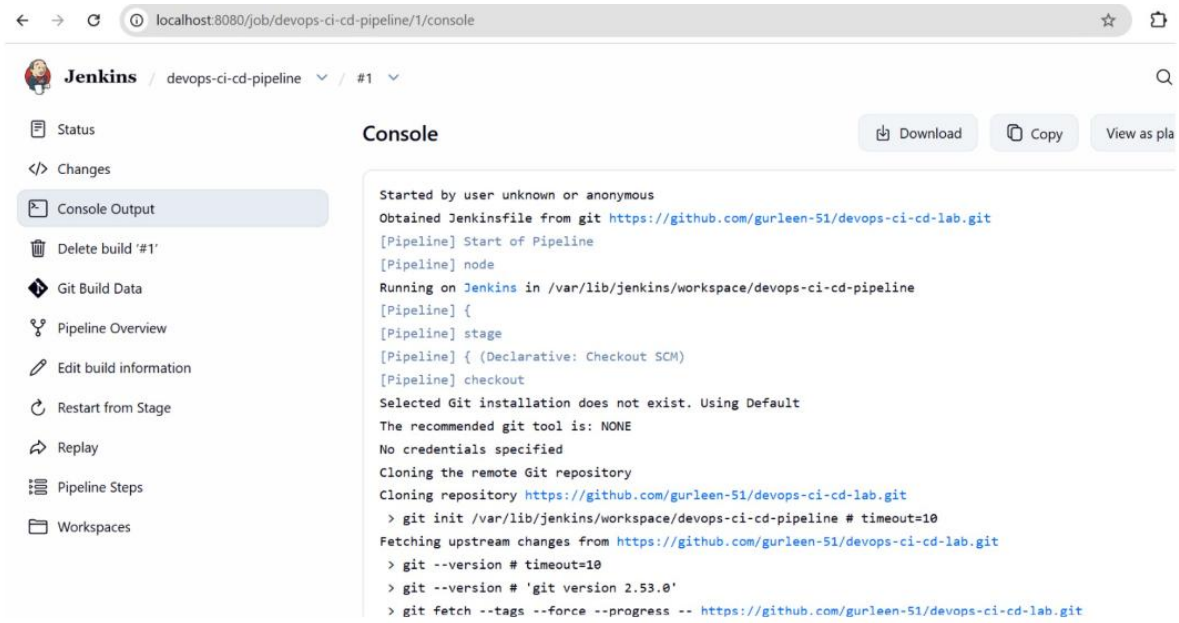
Jenkins automatically performed:

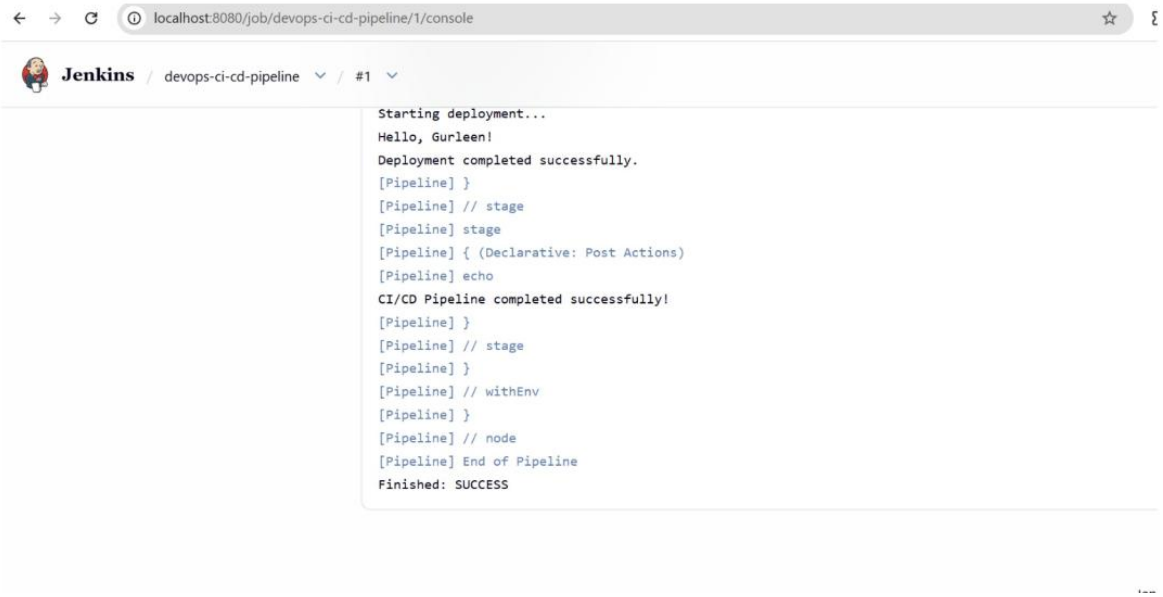
GitHub Change
↓
Jenkins Trigger
↓
Build
↓
Test
↓
Deploy
↓
SUCCESS

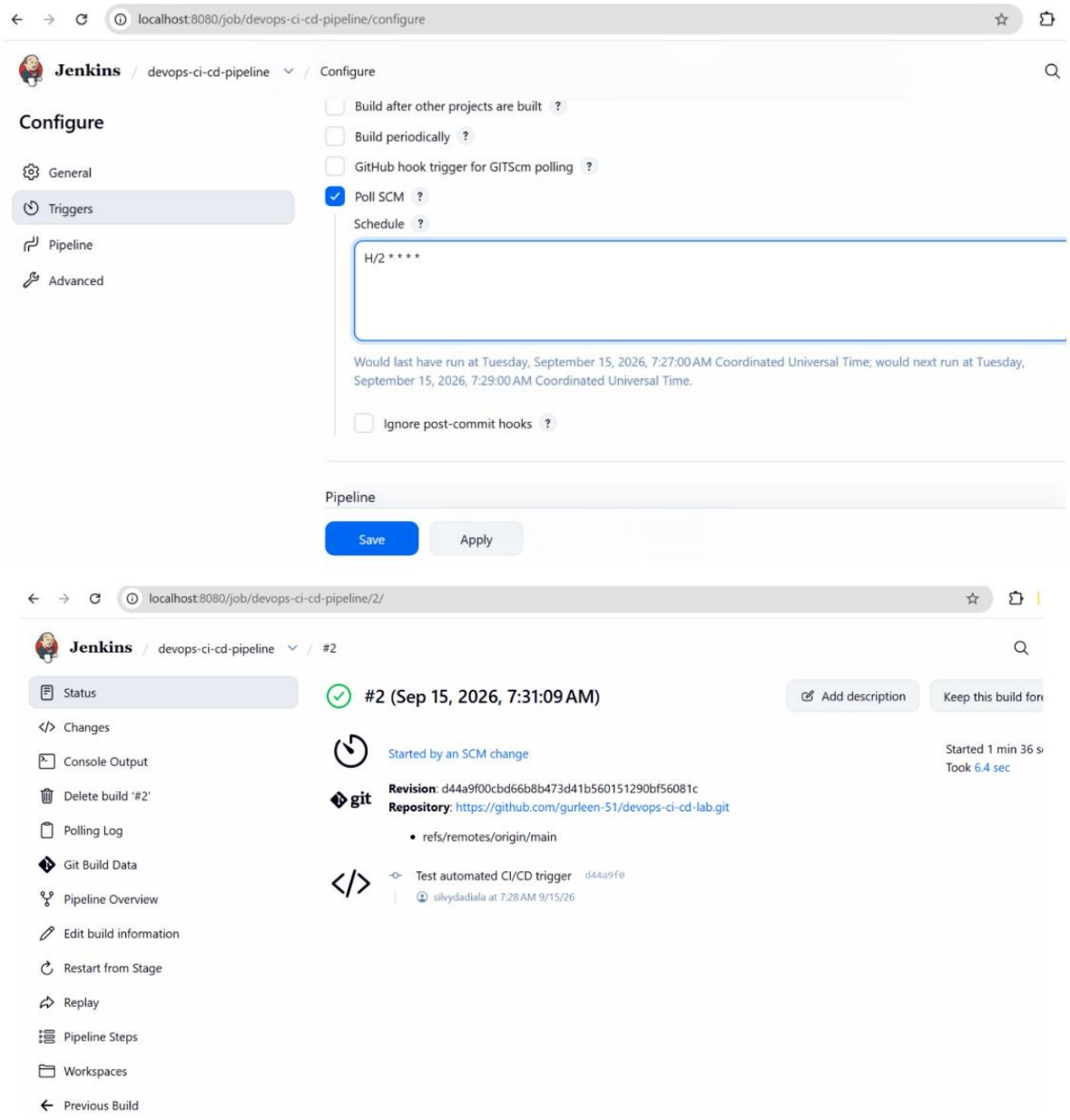

Jenkins
New Item
Build History
Build Queue
No builds in the queue.
Build Executor Status
0/2
Welcome to Jenkins!
This page is where your Jenkins jobs will be displayed. To get started, you can s
builds or start building a software project.
Start building your software project
Create a job
Set up a distributed build
Set up an agent
Configure a cloud
gurleen
Theme
Light
My Views
Account
Appearance
Preferences
Security
Experiments


localhost:8080/view/all/newJob
Jenkins / All / New Item
New Item
Enter an item name
devops-ci-cd-pipeline
Select an item type
Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
OK


localhost:8080/job/devops-ci-cd-pipeline/configure
Jenkins / devops-ci-cd-pipeline / Configure
Configure
General
Triggers
Pipeline
Advanced
Repositories
Repository URL
https://github.com/gurleen-51/devops-ci-cd-lab.git
Credentials
- none -
Add
Advanced
Add Repository
Branches to build
Save
Apply


localhost:8080/job/devops-ci-cd-pipeline/1/console
Jenkins / devops-ci-cd-pipeline / #1
Status
Changes
Console Output
Delete build '#1'
Git Build Data
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Console
Download
Copy
View as pla
Started by user unknown or anonymous
Obtained Jenkinsfile from git https://github.com/gurleen-51/devops-ci-cd-lab.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/devops-ci-cd-pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/gurleen-51/devops-ci-cd-lab.git
> git init /var/lib/jenkins/workspace/devops-ci-cd-pipeline # timeout=10
Fetching upstream changes from https://github.com/gurleen-51/devops-ci-cd-lab.git
> git --version # timeout=10
> git --version # 'git version 2.53.0'
> git fetch --tags --force --progress -- https://github.com/gurleen-51/devops-ci-cd-lab.git


localhost:8080/job/devops-ci-cd-pipeline/1/console
Jenkins / devops-ci-cd-pipeline / #1
Starting deployment...
Hello, Gurleen!
Deployment completed successfully.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
CI/CD Pipeline completed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS



Jenkins successfully integrated with GitHub and automatically executed the pipeline using the Jenkinsfile stored in the repository.